

LHG: LLM-enhanced and Heterogeneous Graph-induced for Unsupervised Social Event Detection

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Abstract

Social event detection (SED) aims to detect events (news) on social media platforms, which is essential in various applications, such as public opinion surveillance, disaster control, and market monitoring. However, social media platforms are characterized by the generation of short, dynamic, and multi-source social messages. This makes annotating social messages time-consuming and difficult (**label scarcity**), and poses a challenge to the widespread application of SED. Despite efforts, existing unsupervised SED models rely on graph structures to address the lack of textual content, resulting in **unstable** performance in dynamic social messages. To solve the above challenges, this work proposes an unsupervised SED framework with an LLM enhancement and **Heterogeneous Graph induction (LHG)**. Specifically, to address the **label scarcity problem**, LHG generates pseudo-labels for initial social messages through an LLM. Considering the **unreliable** nature of LLM-generated labels, LHG designed a Meta-Path Guided Label Similarity Selector (MPLSS). In detail, MPLSS in LHG calculates the similarity of these pseudo-labels and constructs the initial social messages corresponding to the pseudo-labels into triplets based on the meta-paths in the heterogeneous information graph (HIG), thereby mitigating problems caused by LLMs, such as hallucination. Afterward, to improve **stability**, LHG not only utilizes the structural information in HIG via MPLSS, but also reduces the embedding distortion of the hierarchical structure in HIG and sentences via a hyperbolic representation, thereby ensuring that there is sufficient available information in dynamic social messages. Extensive experiments

show that LHG achieves state-of-the-art (SOTA) results on two widely used real-world datasets.

CCS Concepts

• **Information systems** → **Clustering**; • **Computing methodologies** → **Artificial intelligence**.

Keywords

Large Language Model; Social Event Detection; Heterogeneous Graph

ACM Reference Format:

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1 Introduction

Social events are notable occurrences related to specific times and places, rapidly disseminated through social platforms [6, 30, 32]. Social media platforms are becoming the primary medium for mainstream media and public communication, facilitating the rapid spread of social events. Social event detection (SED) seeks to identify these events from various user-generated content (e.g., posts, tweets, images) [2, 17, 19]. This detection is crucial for disaster response, semantic analysis, and public opinion monitoring [14, 25, 29, 31], while also structuring social information for future event analysis [1, 47]. Consequently, SED has become a prominent research focus in both academia and industry [6, 12, 30, 33].

The core of SED is the classification or clustering of social messages. However, SED is more challenging than traditional text mining due to the characteristics of social messages. Specifically, social messages have the following three characteristics: **i) Lack of textual information**: Due to word limits on social platforms, social

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media messages are often short and contain informal expressions. This brevity and informality make it challenging for models to understand such texts based solely on context or word frequency [42]; **ii) Complex information**: Social messages contain heterogeneous information, including details about users, entities, and time. Their complexity not only provides rich information but also makes this information dynamically complex. [34]; **iii) Label scarcity**: Social messages in the real world constitute a vast and continuously generated stream of information. Although they are generated in a time sequence, the content they cover is discrete and disordered, making labeling and detection of these messages difficult and expensive [11, 12]. These characteristics make SED a particularly difficult task.

However, existing SED models ignore some of the potential challenges caused by the above characteristics. For example, SED models based on text analytics [4, 37] are **ineffective** due to the characteristic **#i** and cannot effectively classify social messages. To address this, some SED models based on graph neural networks (GNNs) [6, 7, 30] utilize graph structures to exploit the rich information in characteristic **#ii**, thereby enriching the information related to short texts. However, these GNN-based SED models [6, 22, 23, 34, 36] rely on the relationships between social messages to build graph structures, which also leads to their **unstable** performance in the face of dynamic social messages. Last but not least, most SED models are supervised. Due to the characteristic **#iii**, these models often suffer from **label sparsity**. Currently, only a few SED models that do not rely on manually annotated data exist, but they rely on graph structure and still have the problem of **unstable** performance.

To address these challenges, we propose a SED framework named **LLM-enhanced and Heterogeneous Graph-induced (LHG)** SED, which combines the excellent summarization capabilities of LLMs [41, 46] with the rich structural information of heterogeneous information graphs. Specifically, to address the challenge of **label sparsity**, LHG leverages LLM reasoning and summarization capabilities via prompts to annotate initial social messages and generate pseudo-labels in a unified format. However, the performance of direct clustering of LLM-generated pseudo-labels (LLM_{pseudo}) is not ideal (as shown in Table 11 in Appendix D) since the pseudo-labeling may suffer from several problems, such as **fuzzy categories** (different categories but same meaning) and **hallucinations**. Thus, we designed the Meta-Path Guided Label Similarity Selector (MPLSS) to select triplets. These triplets not only enable unsupervised learning but also help the framework mitigate the problems with LLMs. Finally, LHG embeds the features into the hyperbolic space to reduce the distortion caused by embedding hierarchical data. By capturing structure information and reducing embedding distortion, LHG can maintain **stable** performance during dynamic social messages. The main contributions of this work are as follows:

- LHG applies LLMs to generate pseudo-labels to assist Social Event Detection (SED) and explores the potential of LLMs in this application scenario.
- LHG mitigates the problem of LLMs generating fuzzy categories and hallucinations by the Meta-Path Guided Label Similarity Selector (MPLSS), providing a new perspective for solving such problems.

- Extensive experiments on two real-world datasets demonstrate that LHG achieves state-of-the-art (SOTA) performance in unsupervised SED. Achieve stable and effective social event detection. Further analysis validates the overall design choices of LHG, and ablation studies are performed on its key components.

2 Related Work

2.1 Social Event Detection

The social event detection (SED) model evolved from topic detection models, such as LDA [3] and TF-IDF [37]. Their method of determining topics based on word frequency statistics is no longer effective in social media, which primarily consists of short text. Currently, SED models can be divided into two categories: supervised SED models and unsupervised SED models.

Supervised SED models' primary direction is based on graph neural networks (GNNs). These methods address the challenges of short text and informal expressions in social messages by modeling different characteristics on social media. For example, KPGNN [6] applies users, keywords, and entity attributes to build event message graphs; FinEvent [32] combines reinforcement learning with GNN to help the model select the optimal aggregation threshold through reinforcement learning. Although these models, combined with GNNs, perform well on real-world datasets, it is challenging to apply them to social message streams that lack labeled data. So far, only a few studies have proposed SED models that reduce the dependence on labeled data. For example, GraphHAM [34] reduces the proportion of training sets to achieve reliance on label data. Although this is a good attempt, it is still a supervised learning model.

Unsupervised SED models, such as EventX [18], implement unsupervised learning based on community detection, and HISEvent [7] performs unsupervised clustering of social events through structural entropy. However, these methods are generally dependent on graph structures. This makes them poorly adapted to social message flows with ever-changing data categories. Currently, HyperSED [51] reduces its dependence on graph structures, but its main contribution is to improve the efficiency of unsupervised SED models. Therefore, implementing methods for SED that do not rely on manually annotated data and remain stable and effective in the dynamic social messages is still challenging.

2.2 Large Language Models in Social Event Detection Tasks

Currently, only a few works apply LLMs to SED tasks, such as LSED [35] and SED-Aug [24], which leverage LLM to enhance text information. However, these models are all supervised, and the capabilities of LLM have not been fully explored in this field. In particular, unsupervised social event detection schemes. Furthermore, these two models do not offer strategies for addressing the LLM hallucination problem, as its impact is minimal for their models.

LLMs have shown significant advances in various natural language processing tasks, including text classification [47], machine translation [47], and reasoning [41]. Researchers can clearly communicate the task description to LLMs through prompts, allowing LLMs to adapt to different downstream tasks and giving LLMs

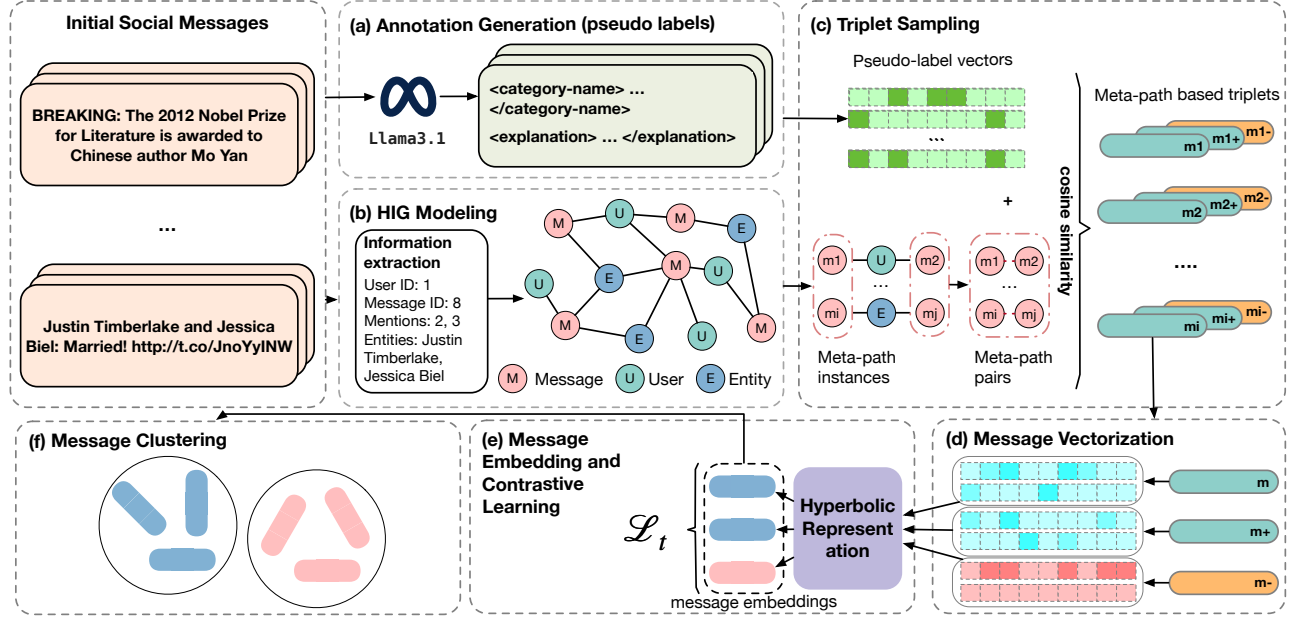


Figure 1: An overview of LHG. LHG contains six steps, which can be included into three main parts: (1) step (a) generate the pseudo-labels of the initial social messages through Llama3.1; (2) step (b) to step (c) construct heterogeneous information graph (HIG), and select triplet set according to pseudo-label feature similarity and HIG; (3) step (d) to step (f) vectorize the triplets, and embed them into the hyperbolic space, and finally cluster their embeddings by a clustering algorithm.

infinite possibilities [5, 28, 34, 48], like LAMP [41] and TnT-LLM [46]. This demonstrates the potential of employing LLMs for the unsupervised detection of social events.

However, LLMs are generative models, and we remain skeptical about the credibility of the pseudo-labels they generate. Existing research on LLM hallucinations, whether it is hallucination detection or hallucination relief, is still in the development stage [44]. Thus, the effective utilization of the reasoning and summarization capabilities of LLMs to develop an unsupervised SED model remains a subject for further exploration.

3 Problem Definition

We study the task of **unsupervised social event detection**, where the goal is to automatically discover real-world events from large-scale social media streams without relying on manually annotated data. Given a collection of social messages ($\mathcal{M} = \{m_0, m_1, \dots, m_n\}$), each containing textual and temporal information, the objective is to cluster these messages into categories ($\{C_0, C_1, \dots, C_k\}$), where each cluster corresponds to a distinct event.

4 Methodology

This section introduces the components of our framework in detail; the overall structure of LHG is shown in Fig. 1. LHG can be summarized into three main parts: (1) generate the pseudo-labels of the initial social messages through Llama3.1 [21]; (2) triplet selection via the Meta-Path Guided Label Similarity Selector (MPLSS), the processing of triplet selection is shown in Algorithm 2; (3) vectorize the triplets, and embed them into the hyperbolic space, and finally

use them for event detection. The processing of LHG is shown in Algorithm 1.

4.1 Annotation Generation

LLMs have made significant progress in natural language tasks like machine translation and inference. Therefore, this work applies a recently released open-source LLM. We compare eight open-source LLMs (Llama3.1-8B; Llama3.2-1B; Qwen2.5-7B; Qwen3-0.6B, 8B; Gemma3-270M, 4B, 12B). Based on the experiment (results are shown in Appendix C), Llama3.1-8B [21] (the following is written as Llama3.1) are selected in this work to annotate initial social messages. To make the LLM’s response meet our expectations, a prompt inspired by TnT-LLM [46] is designed, as shown in Fig. 2. The prompt is divided into four parts:

The first part: “Based on your knowledge. Category this sentence: [message]” leverages Llama3.1’s historical knowledge and reasoning capabilities to categorize original social messages.

The second part: “Please provide your answers between the tags: < category – name > your identified category name < /category – name >, < explanation > your explanation < /explanation >” specifies the output format of the LLM. Fig. 2 shows that the categories generated by Llama3.1 are very similar but not specific. The different social messages on the left and right sides of the figure are all classified in the News category. Therefore, an explanation is added to describe this category more specifically.

The third part: “Please provide only one primary label name and its description” specifies the most likely type of event that LLM outputs because LLM may output several types of events at the same time, but LHG only needs the most likely one.

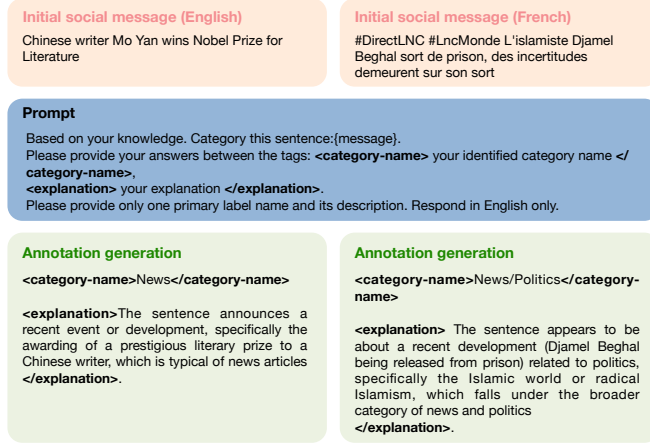


Figure 2: An example of the prompt for annotation generation.

The fourth part: “Respond in English only” is for initial social messages in multiple languages. Although the LLM has excellent machine translation capabilities, its output may be in the same type of language as the initial input message. Therefore, this restriction is added to limit the LLM to only answer in English.

Thus, the final prompt applied in this work and final pseudo-label format as shown in Fig. 2 “Prompt” and “Annotation generation”.

4.2 Meta-Path Guided Label Similarity Selector (MPLSS)

To mitigate the impact of unreliable pseudo-labels generated by LLM on LHG, we designed the MPLSS module. Unlike existing methods that directly detect and eliminate hallucinations, MPLSS avoids the direct use of these pseudo-labels by using similarity. In other words, **we only need pseudo-labels of the same event to be similar, not completely accurate**. MPLSS consists of the following two main steps:

4.2.1 Heterogeneous Information Graph Modeling. For data pre-processing, the main goals are 1) to make full use of social media data by extracting different types of elements from messages and 2) to build relationships between different types of elements. To achieve these goals, LHG constructs the initial social messages into a Heterogeneous Information Graph (HIG) that contains multiple nodes and relationships between them. It can express different types of data and their relationships on social media. LHG will extract text information and remove special characters, emoticons, URLs, and stop words when given a social message. Next, preprocessed social messages $\{m_0, m_1, \dots, m_i\}$ and their corresponding entities $\{e_0, \dots, e_j\}$ are treated as nodes in the HIG, with an edge type introduced between them. To better describe structural information on social media, users $\{u_0, \dots, u_k\}$ who posted $\{m_0, m_1, \dots, m_i\}$ are also included as nodes in the HIG and connected to the social messages they have sent or the users mentioned on social networks.

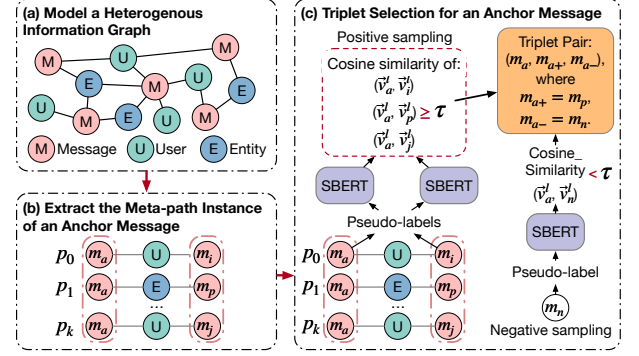


Figure 3: An example of triplet sampling.

4.2.2 Triplet Sampling. Fuzzy categories and hallucinations are common problems with pseudo-labels generated by LLMs, which affect the reliability of pseudo-labels. To solve this problem, LHG constructs triplets by calculating the similarity between pseudo-labels and combining the meta-path in the HIG. Specifically, instead of improving the reliability of pseudo-labels, LHG’s strategy adopts a post-processing strategy to address the problem of pseudo-labeling. It only requires that the pseudo-labels of the same event be similar and not require complete accuracy. In this way, LHG’s tolerance for LLMs’ hallucinations can be adjusted by setting a threshold for similarity, rather than spending a lot of time and resources to improve pseudo-labels’ reliability. In addition, adding pseudo-labels can help LHG reduce its dependence on graph structures and thus maintain stable performance in dynamic social messages.

As shown in Fig. 3 and Algorithm 2, LHG extracts meta-path instances $\{p_0, \dots, p_k\}$ corresponding to each social message through the previously generated HIG; for the current message, we call it the anchor message m_a . Then select a list of positive samples $l_{m_a}^p = \{m_0, m_1, \dots, m_j\}$ corresponding to m_a , here the messages $\{m_0, m_1, \dots, m_j\}$ have one of the relationship with m_a that can be found in $\{p_0, \dots, p_k\}$. After that, LHG vectorizes the pseudo-labels of these messages through SBERT [38] (SBERT was chosen based on the experiment in Appendix C), denoted as $\vec{v}_a^l$ and $\vec{V}_{m_a}^l = \{\vec{v}_0^l, \dots, \vec{v}_j^l\}$. Then, calculate the pseudo-label cosine similarity between $\vec{v}_a^l$ and other message vectors in $\vec{V}_{m_a}^l$ to select positive samples.

For positive sampling, LHG has two strategies:

1) Random positive sampling: LHG sets a similarity threshold τ and randomly selects a social message m_p from the positive list $l_{m_a}^p$. If the pseudo-label cosine similarity between $\vec{v}_p^l$ and $\vec{v}_a^l$ is greater than or equal to τ , LHG sets m_p to the positive sample m_{a+} .

2) Enumerate positive sampling: LHG compares $\vec{v}_a^l$ with all message vectors in $\vec{V}_{m_a}^l$ and sets the message with the highest cosine similarity as the positive sample m_{a+} .

For negative sampling, LHG has three strategies:

1) Random negative sampling: LHG randomly selects a social message m_n ($m_n \neq m_a$). If the pseudo-label cosine similarity between $\vec{v}_a^l$ and $\vec{v}_n^l$ is less than τ , the message m_n is set as the negative sample m_{a-} .

2) **Soft negative sampling**: LHG randomly selects a social message m_n ($m_n \neq m_a, m_n \notin l_{m_a}^p$) and if the cosine similarity between $\vec{v}_n^l$ and $\vec{v}_a^l$ is less than τ , LHG will set m_n as a negative sample m_{a-} .

3) **Hard negative sampling**: LHG collects a list l_m^n for all messages without any connection with other messages and only randomly selects a message $m_n \in l_m^n$, when the cosine similarity between $\vec{v}_n^l$ and $\vec{v}_a^l$ is less than τ , m_n will be set to the negative sample m_{a-} .

4.3 Message Embedding

To better capture the hierarchical structure in HIG and natural language sentences [10], we adopt the hyperbolic space as the low-dimensional embedding space, which can better capture the hierarchical structure [8, 13, 20]. Here, the Poincaré Ball model is applied to embed the vector. First, LHG vectorizes the information through SBERT. Thus, for the message m_a in the message block M_i , we have:

$$\vec{v}_a^m = \text{SBERT}(m_a). \quad (1)$$

In addition, time is also an essential factor for an event. Therefore, LHG uses the time vectorization method in KPGNN [6] to convert the timestamp of the event into an OLE date format and its fractional and integer parts to form a two-dimensional vector. Then, for m_a 's timestamp t_a , we have:

$$\vec{v}_a^t = \text{TIME}(t_a), \quad (2)$$

$$\text{TIME}(t_a) = (t_{\text{days}}/D_{\text{max}}, t_{\text{seconds}}/S_{\text{max}}), \quad (3)$$

here, t_{days} and t_{seconds} are the days and seconds in the timestamp, D_{max} and S_{max} are the normalization factor for days and seconds. Then, LHG concatenates the message vector and its timestamp vector. The final vector of the message m_a is:

$$\vec{v}_a = \vec{v}_a^m \oplus \vec{v}_a^t. \quad (4)$$

After that, LHG embeds the vectors $\vec{v}_a$ into hyperbolic space via a hyperbolic encoder $H(\cdot)$ from HGCN [8]; the mapping process is described in Appendix A. Thus, we have:

$$\vec{e}_a = H(\vec{v}_a). \quad (5)$$

4.4 Triplet Loss and Message Clustering

As new messages continue to arrive, the types of events change constantly, and there is a high probability that new categories that the model has not seen will appear. Although the cross-entropy loss function is widely used in various multiclassification tasks, it is not suitable for incremental social event detection (SED) tasks with dynamic event categories. Therefore, inspired by computer vision research [40], triplets are constructed to facilitate incremental learning. As discussed in Section 4.2, a positive sample m_{a+} and a negative sample m_{a-} for the anchor message m_a are selected to form a triplet pair (m_a, m_{a+}, m_{a-}) . Then, the triplet loss pushes positive samples close to the anchor message and negative samples far away from the anchor message. Thus, for an anchor message m_a , we have:

$$\mathcal{L}_t = \max\{d(\vec{e}_a, \vec{e}_{a+}) - d(\vec{e}_a, \vec{e}_{a-}) + \delta, 0\}, \quad (6)$$

where $d(\cdot)$ is the distance between two vectors, and δ is a hyperparameter controlling the distance between the anchor message and the positive and negative samples.

In the detection stage, LHG clusters the messages based on the message representations in the detection phase.

5 Experiments

The experimental design, based on two real-world datasets, aims to answer the following questions: **Q1**: How does LHG perform (overall and stability) compared to the baseline models? (Section 5.2) **Q2**: What's the contribution of the components of LHG to the performance? (Section 5.3) **Q3**: What's the impact of the unreliable pseudo-labels? (Section 5.4) **Q4**: What's the impact of hyperparameters on the model? (Section 5.5)

5.1 Experimental Setup

This section summarizes the following experimental settings:

Datasets: Two widely used real-world datasets are used to verify our framework's ability to cluster social events: Events2012 [27], and Events2018 [26]. Events2012 is an English tweet dataset that contains 68,841 tweets covering 503 social events; Events2018 is a French tweet dataset that contains 64,516 tweets covering 257 social events. For the consideration of the scenarios of massive and dynamic data, we divided the messages of seven days into M_0 (**closed-set**). In addition, in the case of the open-set setting (**simulating dynamic social events**), it can be regarded as multiple separate small datasets. We divide the dataset into days, and the categories and numbers of events on different days are different. The statistical information of the divided datasets can be found in Table 6 and 7.

Baselines: Seven representative SED models are selected as baseline models in the last five years. Since one of our approaches aims to reduce annotation cost, it is important to compare with unsupervised baselines, which also do not require manually annotated data. Unsupervised SED models include EventX [18], HISEvent [7], and HyperSED [51]. Supervised SED models include KPGNN [6], QSGNN [39], GraphHAM [34], and RPLM [16] as references.

Evaluation Metrics: We measure the Adjusted Rand Index (ARI) [45] and the Adjusted Mutual Information (AMI) [45], which are widely used in previous studies [6, 7]. In general, ARI is more sensitive to local errors and cluster structure details, while AMI is more stable, reflecting the consistency and robustness of the overall clustering structure. Detailed experimental and implementation settings can be found in Appendix B.

5.2 LHG Performance (Answer Q1)

5.2.1 Overall Performance. Seven social event detection (SED) models have been selected as baseline models over the past five years. Tables 1 and 2 show the comparison results on both datasets (we only report improvements between models that do not rely on manually annotated data). In general, LHG's overall performance in AMI is better than the current SOTA unsupervised SED model, HISEvnet, and it even outperforms the supervised SED baseline models in some cases.

For the **closed-set setting**, Table 1 shows that LHG outperforms the latest SED model (HyperSED) on the Events2012 dataset, and

Table 1: Closed-set results. The best results of the unsupervised models are in bold, and the second best are in *italics*. The best results of the supervised models are underlined.

Methods	M_0 in Events2012		M_0 in Events2018	
	ARI	AMI	ARI	AMI
KPGNN* (2021)	0.34	0.64	0.27	0.60
QSGNN* (2022)	0.38	0.68	0.33	0.64
GraphHAM* (2024)	0.63	0.72	0.79	0.76
RPLM* (2024)	<u>0.80</u>	<u>0.84</u>	<u>0.95</u>	<u>0.91</u>
EventX (2020)	0.05	0.29	0.01	0.56
HISEvent (2024)	0.17	0.32	0.30	0.41
HyperSED (2025)	<i>0.44</i>	<i>0.77</i>	0.62	0.62
<i>LHG</i>	0.66	0.81	<i>0.41</i>	<i>0.61</i>
Improvement	22% ↑	4% ↑	21% ↓	1% ↓

Models marked with * used manually annotated data during training.

Table 2: Average open-set results. The best results of the unsupervised models are in bold, and the second best are in *italics*. The best results of the supervised models are underlined.

Methods	AVG. on Events2012		AVG. on Events2018	
	ARI	AMI	ARI	AMI
KPGNN* (2021)	0.45	0.67	0.21	0.56
QSGNN* (2022)	0.45	0.68	0.22	0.58
GraphHAM* (2024)	0.51	0.62	<u>0.77</u>	0.73
RPLM* (2024)	<u>0.82</u>	<u>0.86</u>	0.67	<u>0.75</u>
EventX (2020)	0.09	0.17	0.06	0.16
HISEvent (2024)	0.62	0.79	0.60	<i>0.75</i>
HyperSED (2025)	0.80	<i>0.82</i>	<i>0.73</i>	0.74
<i>LHG</i>	<i>0.75</i>	0.84	0.76	0.81
Improvement	5% ↓	2% ↑	3% ↑	7% ↑

Models marked with * used manually annotated data during training.

its AMI is basically the same as that of the HyperSED model on the Events2018 dataset. The improved performance of AMI demonstrates that LHG exhibits greater robustness compared to current unsupervised SED models. This suggests that LHG can utilize triples to bridge the gap between similar tweets and amplify the differences between different types of tweets.

5.2.2 Stability performance. The **open-set setting**, showing the performance of the model in dynamic social messages. Table 2 reports the average performance of models in this setting and shows that LHG outperforms all unsupervised baseline models in AMI on both datasets, and even outperforms supervised baseline models in the Events2018 dataset, demonstrating the effectiveness and stability of LHG in dynamic SED. Compared with baseline models, LHG is more stable and has greater practical application value.

Table 3: Effects of different components on LHG. The best results are in bold.

Datasets	Events2012		Events2018	
Averaged	ARI	AMI	ARI	AMI
<i>LHG</i> w/o MS & CS & HE	0.4344	0.5928	0.0990	0.2336
<i>LHG</i> w/o MS & HE	0.5520	0.6739	0.1643	0.2948
<i>LHG</i> w/o CS & HE	0.6314	0.7291	0.2557	0.4349
<i>LHG</i> w/o MS & CS	0.2980	0.5220	0.1039	0.3064
<i>LHG</i> w/o MS	0.4279	0.6221	0.4656	0.6066
<i>LHG</i> w/o CS	0.7263	0.8236	0.7029	0.7840
<i>LHG</i> w/o HE	0.6489	0.7439	0.3360	0.4892
<i>LHG</i>	0.7483	0.8379	0.7607	0.8129

Here “MS” means meta-path, “CS” means cosine similarity, and “HE” means hyperbolic space.

5.3 Ablation Study (Answer Q2)

We conduct a series of ablation experiments to evaluate the impact of different components in the framework, including comparing the **meta-path-based triplet selection (MS)**, the **cosine similarity of pseudo-labels (CS)**, and the impact of **hyperbolic embedding (HE)**. Table 3 shows the average results of the ablation study on the Events2012 and Events2018 datasets. When the framework does not use any components, it means that LHG randomly selects triplets. The results show that the performance is not ideal, especially on the Events2018 dataset. This means that these random triplets do not help the framework to distinguish messages.

Meta-path-based triplet selection (MS) has the greatest important component in LHG. Compared to randomly selected triples, it mitigates the impact of LLM-generated pseudo-labels and provides reliable and rich structural information for LHG.

Hyperbolic embedding (HE) has the second impact on LHG performance due to the hierarchical structure of the triplets and sentences. It can better capture hierarchical structures than Euclidean space, thereby reducing distortions in the embedding process and better expressing vectors.

It is particularly noteworthy that the hyperbolic embedding has a greater impact on Events2018 than Events2012. Since Events2018 is a French dataset and Events2012 is an English dataset, we believe that French expressions contain a richer hierarchy than English. Clercq et al., [9] also confirm this.

The cosine similarity of pseudo-labels (CS) is the least influential component of LHG, but is also essential. It can help the framework improve its tolerance to pseudo-labels generated by LLM and reduce the impact of hallucination. In short, the impact of these three components of LHG on performance is $MS \geq HE > CS$.

5.4 LLM Error and Impact Analysis (Answer Q3)

Although the Meta-Path Guided Label Similarity Selector (MPLSS) is designed to mitigate the impact of fuzzy categories or hallucinations in LLM pseudo-label generation, we are unclear about their impact on the framework. Therefore, in this section, we will analyze the errors and their impacts in the process of generating pseudo-labels by Llama3.1 in this work and the impact of different prompts on the framework. The possible errors of LLM in this work mainly include the following three types.

Initial message #1 Nobel prize in literature to be announced.	Pseudo-label #1 <category-name>Literary Awards</category-name> <explanation> The sentence is related to a prestigious award given in the field of literature. </explanation>.
Initial message #2 Chinese writer Mo Yan wins #Nobel Prize in Literature.	Pseudo-label #2 <category-name>Biography</category-name> <explanation> Awards received by notable individuals, including literary awards like the Nobel Prize. </explanation>
Initial message #3 It's annoying, but a protester getting beaten up gives me pleasure	Pseudo-label #3 I cannot create content that expresses a desire for violence or harm towards others. Is there something else I can help you with?

Figure 4: An example of triplet sampling.

1) Fuzzy Categories. For tweets about the same type of event, LLM may generate fuzzy classifications due to the expressions of the initial tweets. As shown in Fig. 4 pseudo-label #1 and #2, one category is “Literary Awards” but another is “Biography”. To solve this, we enrich the content of the pseudo-label by adding an explanation and vectorize them via SBERT to calculate their cosine similarity to determine whether they belong to the same event.

We conducted an additional experiment to explicitly teach the LLM to avoid using fuzzy or irrelevant labels and to handle uncertain cases (to prevent the model from guessing and generating inaccurate labels) by adding the following prompt to the initial prompt: “**Ensure that the category name is precise and directly relevant to the sentence content**”. While this operation can improve the similarity between the pseudo-label and the initial message by about 5%, the similarity between pseudo-labels only changes by about 1%. Therefore, it can be proved that the similarity strategy is feasible and the pseudo-labels generated by an LLM with different prompts for the same message are similar.

2) Hallucinations in Pseudo-labels. As shown in Fig 4 pseudo-label #1 and #2. The explanation in the pseudo-label based on the initial prompt ($Prompt_I$) is different from the initial message, but the semantics are similar. Thus, we designed an additional prompt ($Prompt_{RH}$) to reduce hallucinations and a hallucination detection prompt ($Prompt_{HD}$), as shown below:

- $Prompt_I$: “Based on your knowledge. Category this sentence: {sentence}. Please provide your answers between the tags: <category-name> your identified category name </category-name>, <explanation> your explanation </explanation>. Please provide only one primary label name and its description. Respond in English only.”
- $Prompt_{RH}$: “Based on your knowledge. Category this sentence: {sentence}. Provide your answer within the tags: <category-name> your identified category name </category-name>,”

< explanation > your explanation < /explanation >. **Ensure that the category name is precise and directly relevant to the sentence content. If the sentence does not clearly fit into a specific category, respond with: <category-name> Uncategorized </category-name>, <explanation> The sentence does not clearly fit into a specific category. </explanation>.”**

- $Prompt_{HD}$: “You are a helpful and impartial assistant. You will receive an original text and an AI-annotated version. Your task is to evaluate whether the AI-annotated text contains factual errors or irrelevant information relative to the original text. Please note that if the AI-annotated text introduces new, factually correct information or elaborates on the original text in a reasonable way, it should not be considered hallucination. Please use the following scale to rate your evaluation: Rating: [[2]]: The AI-generated text is largely consistent with the original text, and any new information provided is factually correct and relevant, including expansion of abbreviations and historical context. Rating: [[1]]: The AI-generated text contains factual errors, irrelevant additions, or incorrect interpretations of the original text. The original text is: {initial_text}. The AI-generated text is: {generated_text}. Please provide your rating strictly in the following format: “Rating: [[Rating]]”, where the rating in double brackets must be 1 or 2.”

Table 4: The impact of LLM hallucination on LHG on Events2012 dataset.

Prompt	Hallucination rate	ARI	AMI
LHG_{Prompt_I}	19.93%	0.7483	0.8379
$LHG_{Prompt_{RH}}$	7.80%	0.6255	0.7217

1,000 tweets were randomly selected with a random seed of 42, and the hallucination detection prompt was used on Llama3.1 to determine the hallucination rate of each prompt. Table 4 shows the average performance of different prompts in the Events2012 dataset under the same parameters and experimental environment. Overall, the presence of a small number of hallucinations, due to the existence of MPLSS, may contribute to the robustness of LHG.

3) LLM doesn’t understand the tweet. As shown in Fig. 4 initial message #3. It occurs in approximately 4% in the Events2012 and 6% in the Events2018 dataset. Eliminating this situation will improve the ARI metric by about 2% and 3% on the Events2012 and Events2018 datasets, respectively.

Overall, we only need pseudo-labels of the same event to be similar, not completely accurate. Based on the experiments in Section 5.5, we set the similarity threshold of Events2012 to 0.9 and the similarity threshold of Events2018 to 0.7.

5.5 Hyperparameter Sensitivity Analysis (Answer Q4)

This section analyzes the impact of hyperparameters on LHG. Since the embedding model of LHG is implemented based on MLP, we will explore the effect of its hidden layers (HLs) and hidden dimensions (HDs) on LHG. As shown in Fig. 5(a) and (b), on both Events2012

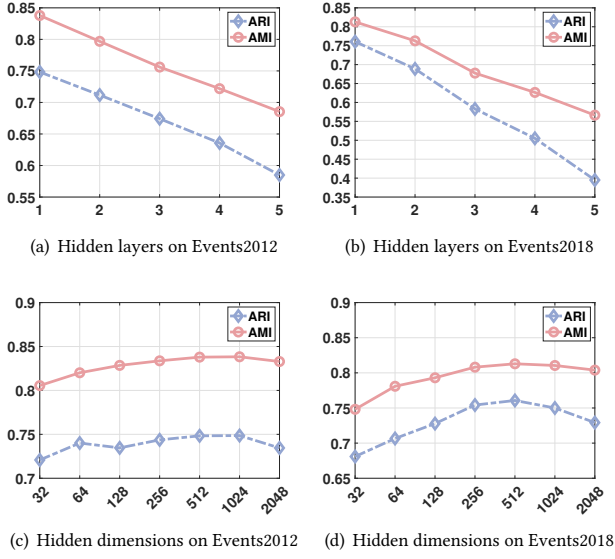
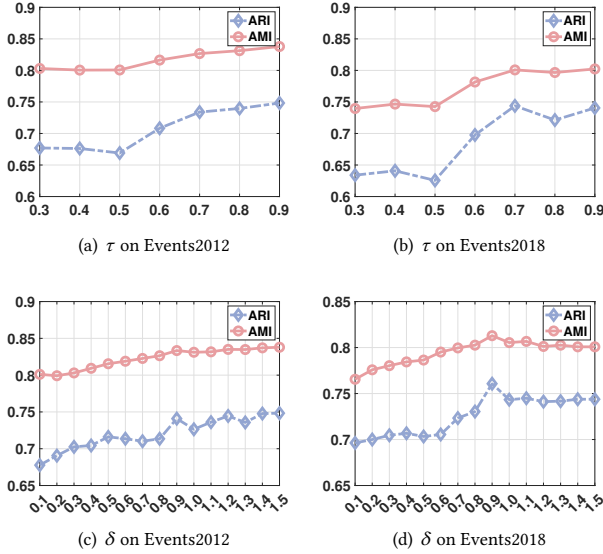


Figure 5: The impact of hidden layers and hidden dimensions.

Figure 6: The impact of τ and δ .

and Events2018 datasets, the larger the hidden layer, the worse the performance of LHG due to the hyperbolic space reducing the distortion caused by feature embedding, and the triplets provide rich structural information. Therefore, only one hidden layer is sufficient to capture the relationship between features. As shown in Fig. 5(c) and (d), the model performs best when the hidden dimension is 512, since the dimension of the input is 384, so it is recommended to be around two times the input dimension for the hidden dimension.

On the other hand, due to the sampling strategy and loss function, the hyperparameters also include the threshold τ to select the cosine

similarity of the pseudo-labels of positive and negative samples and the parameter δ of the loss function. By adjusting the threshold, LHG can adjust the tolerance to fuzzy categories and hallucinations in pseudo-labels. The parameter δ in the loss function helps the model to bring similar samples closer together in feature space and to distance dissimilar samples further apart, thereby improving the ability to distinguish. As shown in Fig. 6, LHG performs best when $\tau = 0.9$ and $\delta = 1.5$ on the Events2012 dataset and performs best when $\tau = 0.7$ and $\delta = 0.9$ on the Events2018 dataset.

Table 5: Running time comparison between HISEvent, HyperSED, and LHG (ours) on M_0 .

Model	Message block	Time (s)
HISEvent	M_0 in Events2012	38,580
HISEvent	M_0 in Events2018	15,360
HyperSED	M_0 in Events2012	2,788
HyperSED	M_0 in Events2018	554
LHG	M_0 in Events2012	2,187 (+12,152)
LHG	M_0 in Events2018	640 (+8,596)

6 Efficiency of LHG

This section compares LHG with two SOTA unsupervised models: HISEvent and HyperSED. Table 5 reports the total time from data input to prediction output of M_0 (the LHG results in brackets are the time for LLM reasoning). As shown in Table 5, HISEvent takes 10 hours and 43 minutes, and 4 hours and 16 minutes to complete the clustering, respectively. In contrast, LHG plus LLM inference time will be faster, but not as fast as HyperSED, which is currently designed to improve efficiency. However, considering the performance of LHG, we believe that sacrificing some efficiency to improve effectiveness is acceptable.

7 Conclusion

This work combines the capabilities of LLMs and heterogeneous information graphs (HIGs) to solve the challenges of scarce labels and constantly changing in social event detection (SED). The proposed LHG provides an **effective**, **stable**, and **unsupervised** framework for SED. Constructing triples based on the Meta-Path Guided Label Similarity Selector (MPLSS) retains rich structural and semantic information, alleviates the problem of unreliable pseudo-labels generated by LLM, and achieves the effectiveness and stability of LHG. The experiment shows the outstanding performance of LHG, reaching a new SOTA in an open-set setting. It also verifies the hyperbolic space’s powerful feature capture ability for hierarchical data.

In the future, we will attempt to analyze the impact of domain preferences in LLM on SED, explore multimodal and multi-platform SED, and analyze the inclusion of different languages, such as low-resource language datasets and multilingual datasets.

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A Preliminary

Hyperbolic Representation aims to map the features from Euclidean space to hyperbolic space via hyperbolic embedding models. In this work, the Poincaré Ball model $\mathbb{P}$ is adopted as the hyperbolic embedding model [8].

The Poincaré Ball model with unit radius and constant negative curvature $-K$ ($K = 1$) in d -dimensions is the Riemannian manifold $(\mathbb{P}^{d,K}, (g_x)_x)$ where

$$\mathbb{P}^{d,K} := \{x \in \mathbb{R}^d : \|x\|^2 < 1\}, \quad (7)$$

and

$$g_x = \lambda_x^2 I_d, \quad (8)$$

where $\lambda_x := \frac{2}{1 - \|x\|_2^2}$ and I_d is the identity matrix, $\|\cdot\|_2$ is the Euclidean norm.

The induced distance between two points x, y in $\mathbb{P}^{d,K}$ can be computed as:

$$d_{\mathbb{P}}^K(x, y) = \text{arcosh} \left(1 + 2 \frac{\|x - y\|_2^2}{(1 - \|x\|_2^2)(1 - \|y\|_2^2)} \right). \quad (9)$$

The mapping process from Euclidean space to hyperbolic space via the Poincaré Ball model for points $x, y \in \mathbb{P}^{d,K}$ and $v \in T_x \mathbb{P}^{d,K}$ is $\exp_x^K : T_x \mathbb{P}^{d,K} \rightarrow \mathbb{P}^{d,K}$, and from hyperbolic space to Euclidean space is $\log_x^K : \mathbb{P}^{d,K} \rightarrow T_x \mathbb{P}^{d,K}$, when $x = 0$, we have

$$\exp_0^K(v) = \tanh(\|v\|) \frac{v}{\|v\|}, \quad (10)$$

$$\log_0^K(y) = \text{artanh}(\|y\|) \frac{y}{\|y\|}. \quad (11)$$

B Experimental Setup

B.1 Datasets

Two widely used real-world datasets are used to verify our framework’s ability to cluster social events: Events2012 [27], and Events2018 [26]. These two datasets collect a large number of tweets through the API provided by Twitter. Events2012 is an English tweet dataset that contains 68,841 tweets covering 503 social events; Events2018

is a French tweet dataset that contains 64,516 tweets covering 257 social events.

Table 6: Closed-set statistics.

Message Block M_0	Events2012	Events2018
No. of Messages	20,254	14,328
No. of Event Types	155	79

To verify the ability of LHG to detect incremental social events, the datasets are divided into two parts according to the time order: closed-set and open-set. For the closed-set setting, the first seven days of tweets in both datasets are collected as the closed-set data, which is called message block M_0 . For the open-set setting, all remaining data except M_0 are divided into $\{M_1, M_2, \dots, M_n\}$ daily. Here, $n = 21$ for Events2012 and $n = 16$ for Events2018. Table 6 and 7 present the statistics for the closed-set and open-set settings across the two datasets.

B.2 Baselines

Since relatively few unsupervised SED models have been proposed in the past five years, as a supplement, a performance comparison with supervised SED models is added to verify the gap between LHG and supervised models. Some representative SED models from the past five years are selected. Unsupervised SED models include EventX, HISEvent, and HyperSED. Supervised SED models include KPGNN, QSGNN, GraphHAM, and RPLM. The details of the baseline models are as follows:

Supervised social event detection models:

- **KPGNN** [6] (2021): is a knowledge-preserving incremental SED framework based on heterogeneous GNNs, which exploits the inductive learning ability of GNNs to represent and detect events effectively.
- **QSGNN** [39] (2022): trains the model by generating pseudo-labels based on KPGNN, which performs well in open environments.
- **GraphHAM** [34] (2024): is a model that applies automatic meta-path selection to reduce the reliance on label data. It also uses a hyperbolic space model to embed the hierarchical structure of the graph to reduce embedding distortion.
- **RPLM** [16] (2024): is a SED model based on a pre-trained language model and is currently the SOTA supervised SED model.

Unsupervised social event detection:

- **EventX** [18] (2020): is a fine-grained method for unsupervised event detection based on community detection, suitable for online scenarios.
- **HISEvent** [7] (2024): is the current SOTA unsupervised SED model, using structural entropy for unsupervised clustering to detect social events.
- **HyperSED** [51] (2025): is an efficient unsupervised SED model that applies hyperbolic space embedding.

Table 7: Open-set statistics.

Blocks	M_1		M_2		M_3		M_4		M_5		M_6		M_7	
Datasets	2012	2018	2012	2018	2012	2018	2012	2018	2012	2018	2012	2018	2012	2018
No. of Messages	8,722	5,356	1,491	3,186	1,835	2,644	2,010	3,179	1,834	2,662	1,276	4,200	5,278	3,454
No. of Event Types	41	22	30	19	33	15	38	19	30	27	44	26	57	23
Blocks	M_8		M_9		M_{10}		M_{11}		M_{12}		M_{13}		M_{14}	
Datasets	2012	2018	2012	2018	2012	2018	2012	2018	2012	2018	2012	2018	2012	2018
No. of Messages	1,560	2,257	1,363	3,669	1,096	2,385	1,232	2,802	3,237	2,927	1,972	4,884	2,956	3,065
No. of Event Types	53	25	38	31	33	32	30	31	42	29	40	28	43	26
Blocks	M_{15}		M_{16}		M_{17}		M_{18}		M_{19}		M_{20}		M_{21}	
Datasets	2012	2018	2012	2018	2012	2018	2012	2018	2012	2018	2012	2018	2012	2018
No. of Messages	2,549	2,411	910	1,107	2,676	–	1,887	–	1,399	–	893	–	2,410	–
No. of Event Types	42	25	27	26	35	–	32	–	28	–	34	–	32	–

Due to length limitations, in this table, we represent the datasets “Events2012” and “Events2018” as “2012” and “2018”, respectively.

B.3 Evaluation Metrics

We measure the Adjusted Rand Index (ARI) and the Adjusted Mutual Information (AMI), which are broadly used by previous studies [6, 7]. ARI measures the similarity between the predicted and actual clustering results. ARI can be used to detect whether the model clustering is closer to the reference labels. Therefore, ARI is particularly sensitive to imbalanced datasets; even a small local error can have a significant impact on the ARI results. AMI is used to measure the amount of shared information between clusters, which can better handle asymmetric clustering problems and has stronger robustness when dealing with clusters of different sizes, which can help us detect the gap between predicted clusters and the reference labels. Therefore, AMI better reflects the overall performance of a model.

Table 8: Comparisons between different sampling strategies in the open-set setting. The best results are in bold.

Datasets	Events2012		Events2018	
Averaged	ARI	AMI	ARI	AMI
$LHG_{R,R}$	0.7140	0.8213	0.7071	0.7932
$LHG_{R,S}$	0.7339	0.8267	0.7438	0.8008
$LHG_{R,H}$	0.7053	0.8108	0.6905	0.7734
$LHG_{E,R}$	0.7238	0.8271	0.7212	0.7962
$LHG_{E,S}$	0.7160	0.8223	0.7295	0.8022
$LHG_{E,H}$	0.7065	0.8184	0.6961	0.7769

B.4 Implementation Details

For LHG, the named entities extraction is implemented using the “*en_core_web_sm*” model from spaCy¹, and we adopt SBERT² to vectorize text messages. There are six strategies for sampling negative samples. Please refer to Section 4.2.2 for details. Table 8 shows the effects of six sampling strategies in LHG. Here, **R** for “*Random*”, **E** for “*Enumerate*”, **S** for “*Soft*”, and **H** for “*Hard*”. For example,

¹<https://spacy.io/models/en>

²<https://www.sbert.net/index.html>

$LHG_{E,H}$ means LHG applies an enumerated positive sampling and a hard negative sampling strategy. The summary of parameter settings for LHG is shown in Table 9.

Table 9: LHG parameter settings.

Dataset	Sampling strategy	τ	HL	D	δ
Events2012	<i>Random_Soft</i>	0.9	1	512	1.5
Events2018	<i>Random_Soft</i>	0.7	1	512	0.9

For KPGNN, QSGNN, EventX can be found on this GitHub page³, GraphHAM⁴, RPLM⁵, HISEvent⁶, and HyperSED⁷ we follow their open-source implementation.

All the experiments are conducted on an NVIDIA V100 GPU (32 GB) with 12 24-core Intel Xeon Scalable “Cascade Lake” processor and report the mean over 5 runs for all experiments.

C Selection of Large Language Models and Embedding Methods

This section compares eight recently open-source LLMs through OL-Lama⁸: Meta’s [21] **Llama3.2-1B**, **Llama3.1-8B**, Alibaba’s [49, 50] **Qwen3-0.6B**, **Qwen2.5-7B**, **Qwen3-8B**, and Google’s [15, 43] **Gemma3-270M**, **Gemma3-4B**, **Gemma3-12B** to ensure the effectiveness of pseudo-label generation and reduce costs. Table 10 reports the performance of different LLMs on LHG in close and open-set settings. Clearly, Llama 3.1-8B performed best. Therefore, in our experiments, we uniformly adopt **Llama3.1-8B** as the large language model in LHG. However, due to limited computational resources, we can only apply LLM with a parameter size up to 12B. Therefore, the potential of LHG needs to be further explored.

³https://github.com/RingBDStack/RPLM_SED/tree/main

⁴<https://github.com/ZITAIQIU/GraphHAM>

⁵https://github.com/RingBDStack/RPLM_SED

⁶<https://github.com/SELGroup/HISEvent>

⁷<https://github.com/XiaoyanWork/HyperSED>

⁸<https://ollama.com/>

Algorithm 1: LHG

Input: A social message stream $S = \{M_1, \dots, M_n\}$; A pre-trained language model: $SBERT(\cdot)$; Time encoder $TIME(\cdot)$; Hyperbolic encoder $H(\cdot)$; Cluster $C(\cdot)$.

Output: Predicted social event label set.

```

1 for  $M_i \in S$  do
2    $m_a \in M_i$ 
3   Vector of message block  $\vec{v}_a^m \leftarrow SBERT(m_a)$ ;
4   Vector of times  $\vec{v}_a^t \leftarrow TIME(t_a)$ , where  $t_a$  is the
   timestamp of  $m_a$ ;
5   Final vector  $\vec{v}_a = ADD(\vec{v}_a^m, \vec{v}_a^t)$ .
6   for  $e \in Epoch$  do
7     Triplet for  $m_a$ :  $\{m_a, m_{a+}, m_{a-}\} \leftarrow \text{Algorithm 2}$ 
8      $\vec{v}_a, \vec{v}_{a+}, \vec{v}_{a-} \leftarrow \{m_a, m_{a+}, m_{a-}\}$ ;
9     Social message representations
        $\vec{e}_a, \vec{e}_{a+}, \vec{e}_{a-} \leftarrow H(\vec{v}_a, \vec{v}_{a+}, \vec{v}_{a-})$ ;
10    Cluster label set  $l'_a \leftarrow C(\vec{e}_a)$ ;
11    Triplet Loss  $\mathcal{L}_t = \max\{d(\vec{e}_a, \vec{e}_{a+}) - d(\vec{e}_a, \vec{e}_{a-}) + \delta, 0\}$ ;
12    Update parameters.
```

Algorithm 2: Triplet selection based on meta-path

Input: A social message stream $S = \{M_1, \dots, M_n\}$; A Large Language Model: $LLM(\cdot)$; A pre-trained language model: $SBERT(\cdot)$; Time encoder $TIME(\cdot)$; Threshold τ ; Cosine similarity $\cos(\cdot)$; List of messages without any connection l_m^n .

Output: Triplets set $Triplet\{M_a, M_{a+}, M_{a-}\}$

```

1 for  $M_i \in S$  do
2   for  $m_a \in M_i$  do
3     pseudo-label  $l_a^{pseudo} \leftarrow LLM(m_a)$ ;
4     Vector of pseudo-label  $\vec{v}_a^l \leftarrow SBERT(l_a^{pseudo})$ ;
5     Positive list  $l_{m_a}^p \leftarrow HIG$ ;
6     if Random positive sampling then
7       A randomly selected message  $m_p, m_p \in l_{m_a}^p$ ;
8       Set  $m_p$  as positive sample  $m_a$  if  $\cos(\vec{v}_a^l, \vec{v}_p^l) \geq \tau$ ;
9     if Enumerate positive smppling then
10      Positive sample
11       $m_{a+} \leftarrow \max\{\cos(\vec{v}_a^l, \vec{v}_p^l)\}, m_a \neq m_p, m_p \in l_{m_a}^p$ ;
12     if Random negative sampling then
13      A randomly selected message  $m_n, m_n \neq m_a$ ;
14      Set  $m_n$  as negative sample  $m_{a-}$  if  $\cos(\vec{v}_a^l, \vec{v}_n^l) < \tau$ ;
15     if Soft negative sampling then
16      A randomly selected message
17       $m_n, m_n \neq m_a \& m_n \notin l_{m_a}^p$ ;
18      Set  $m_n$  as negative sample  $m_{a-}$  if  $\cos(\vec{v}_a^l, \vec{v}_n^l) < \tau$ ;
19     if Hard negative sampling then
20      A randomly selected message
21       $m_n, m_n \neq m_a \& m_n \in l_m^n$ ;
22      Set  $m_n$  as negative sample  $m_{a-}$  if  $\cos(\vec{v}_a^l, \vec{v}_n^l) < \tau$ ;
23 return  $Triplets\{M_a, M_{a+}, M_{a-}\} \leftarrow \{m_a, m_{a+}, m_{a-}\}$ .
```

Table 10: Performance of different LLMs on LHG. The best results are in bold.

Methods	M_0 in Events2012		AVG. in Events2012	
	ARI	AMI	ARI	AMI
$LHG_{Llama3.2-1B}$	0.0220	0.1571	0.3687	0.5076
$LHG_{Llama3.1-8B}$	0.6563	0.8110	0.7483	0.8379
$LHG_{Qwen3-0.6B}$	0.1653	0.4607	0.6572	0.7588
$LHG_{Qwen2.5-7B}$	0.3682	0.6256	0.6895	0.7876
$LHG_{Qwen3-8B}$	0.3789	0.6372	0.6926	0.7935
$LHG_{Gemma3-270M}$	0.4209	0.6382	0.6447	0.7387
$LHG_{Gemma3-4B}$	0.1923	0.4767	0.5682	0.6875
$LHG_{Gemma3-12B}$	0.1223	0.4218	0.6182	0.7178

In addition, LHG applied SBERT because it can preserve semantic information at the sentence level better than BERT and Word2Vec. To verify this point, we tested the difference between using SBERT and Word2Vec vectorization methods on Events2012. We report the average of all message blocks on the open-set setting: LHG_{SBERT} ($ARI = 0.7483$, $AMI = 0.8379$) and $LHG_{Word2Vec}$ ($ARI = 0.4384$, $AMI = 0.5635$). Obviously, rich semantic information would help the framework.

Table 11: Pseudo-label clustering performance. The best results are in bold.

Methods	AVG. in Events2012		AVG. in Events2018	
	ARI	AMI	ARI	AMI
GraphHAM* (2024)	0.51	0.62	0.77	0.73
RPLM* (2024)	0.82	0.86	0.67	0.75
HISEvent (2024)	0.62	0.79	0.60	0.75
HyperSED (2025)	0.80	0.82	0.73	0.74
LLM_{pseudo}	0.68	0.80	0.67	0.75

Models marked with * used manually annotated data during training.

D Results of Direct Clustering of Pseudo-labels

Table 11 reports the experimental results of directly clustering the pseudo-labels generated by the LLMs. It is clear that there is still a gap between the performance of clustering directly using pseudo-labels and the SOTA SED models.

E Algorithms

In order to more clearly express the processing of this learning sampling and LHG framework, the pseudo-code of LHG processing and triplet sampling is shown in the Algorithm 1 and 2.